

Project Notes

Standard Costing & Variance Analysis

(Excel-based standard costing and variance analysis model using an illustrative automobile manufacturing case study.)

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1. Project Information

Project Title: Standard Costing & Variance Analysis — Illustrative Automobile Manufacturing Case Study

Company Context: Maruti Suzuki India Limited

Tool: Microsoft Excel

Project Type: Finance & Accounting Portfolio Project

Primary Areas: Cost Accounting, Standard Costing, Variance Analysis, Financial Modelling and Management Reporting

2. Purpose of the Project

The purpose of this project is to demonstrate the practical application of **standard costing and variance analysis** using Microsoft Excel.

The model establishes a standard cost for an illustrative automobile manufacturing scenario and compares it with illustrative actual costs.

The resulting differences are analysed into material, labour and overhead variances to identify the major drivers of the overall cost difference.

The project also includes a reconciliation control and management-style dashboard to present the results clearly.

3. Important Data Disclaimer

Maruti Suzuki India Limited is used as the **company context** for this case study.

Publicly available FY2025–26 company information is used for contextual reference.

However, the detailed standard and actual cost inputs in the model are **illustrative modelling assumptions**.

They are **not**:

- Maruti Suzuki's confidential data
- Maruti Suzuki's internal costing data
- Officially reported Maruti Suzuki product-level costs
- Actual Maruti Suzuki material consumption
- Actual Maruti Suzuki labour hours or rates
- Actual Maruti Suzuki manufacturing overheads

The purpose is to demonstrate the methodology of standard costing rather than reproduce a company's internal cost accounting system.

4. Project Scope

The model covers the following areas:

Standard Costing

- Standard material cost
- Standard labour cost
- Standard overhead cost
- Total standard production cost

Actual Costing

- Actual material cost
- Actual labour cost
- Actual overhead cost
- Total actual production cost

Variance Analysis

- Material price variance
- Material usage variance
- Labour rate variance
- Labour efficiency variance
- Overhead expenditure variance
- Overhead efficiency variance

Reporting

- Variance summary
- Overall reconciliation
- KPI cards
- Dashboard charts
- Key findings

5. Workbook Structure

The Excel workbook contains nine sheets.

Sheet	Purpose
01 Project Info	Project overview, purpose and scope
02 Assumption & Sources	Sources, assumptions and data classification
03 Standard Cost	Standard quantities, rates and standard cost
04 Actual Cost	Actual quantities, rates and actual cost
05 Material Variance	Material price and usage variance
06 Labour Variance	Labour rate and efficiency variance
07 Overhead Variance	Overhead expenditure and efficiency variance
08 Variance Summary	Consolidated variances and reconciliation
09 Dashboard	KPIs, charts and key findings

6. Model Workflow

The model follows a structured analytical process:

Company Context



Sources & Assumptions



Standard Cost



Actual Cost



Material Variance



Labour Variance



Overhead Variance



Variance Summary



Overall Reconciliation



Dashboard & Findings

This structure separates the underlying cost calculations from the final reporting layer.

7. Standard Cost Model

Standard costing establishes an expected or benchmark cost for the planned production output.

The model considers three major manufacturing cost categories:

1. Materials
2. Labour
3. Overheads

The standard cost model provides the benchmark against which illustrative actual costs are compared.

The standard cost is:

₹13,38,75,000

8. Actual Cost Model

The actual cost model contains the illustrative quantities, rates and costs used to represent the actual production scenario.

The total actual cost generated by the model is:

₹13,71,40,000

The difference between the standard cost and actual cost is subsequently analysed through the variance schedules.

9. Material Variance Analysis

Material variance analysis explains the difference between standard material cost and actual material cost.

The project separates the material variance into:

- Material Price Variance
- Material Usage Variance

9.1 Material Price Variance

Material Price Variance measures the effect of paying a price different from the standard price.

Formula

Material Price Variance

$$= \text{Actual Quantity} \times (\text{Standard Price} - \text{Actual Price})$$

A higher actual material price than the standard price results in an adverse price variance under the model's presentation.

Result

₹25,30,000 Adverse

9.2 Material Usage Variance

Material Usage Variance measures the effect of using a quantity different from the standard quantity allowed for the output.

Formula

Material Usage Variance

$$= \text{Standard Price} \times (\text{Standard Quantity} - \text{Actual Quantity})$$

Using more material than the standard quantity results in an adverse usage variance under the model's presentation.

Result

₹5,50,000 Adverse

9.3 Total Material Variance

The total material variance is:

₹30,80,000 Adverse

Component	Variance
Material Price Variance	₹25,30,000 Adverse
Material Usage Variance	₹5,50,000 Adverse
Total Material Variance	₹30,80,000 Adverse

Material cost is therefore the largest contributor to the overall adverse variance.

10. Labour Variance Analysis

Labour variance analysis separates the labour cost difference into:

- Labour Rate Variance
- Labour Efficiency Variance

10.1 Labour Rate Variance

Labour Rate Variance measures the effect of paying a different labour rate from the standard rate.

Formula

Labour Rate Variance

$$= \text{Actual Hours} \times (\text{Standard Rate} - \text{Actual Rate})$$

Result

₹4,35,000 Adverse

10.2 Labour Efficiency Variance

Labour Efficiency Variance measures the effect of using more or fewer labour hours than the standard hours allowed.

Formula

Labour Efficiency Variance

$$= \text{Standard Rate} \times (\text{Standard Hours} - \text{Actual Hours})$$

Result

₹1,50,000 Favourable

10.3 Labour Cost Variance

The resulting labour cost variance is:

₹2,85,000 Adverse

Component	Variance
Labour Rate Variance	₹4,35,000 Adverse
Labour Efficiency Variance	₹1,50,000 Favourable
Labour Cost Variance	₹2,85,000 Adverse

The favourable efficiency variance partially offsets the adverse rate variance.

11. Overhead Variance Analysis

The model analyses manufacturing overhead through:

- Overhead Expenditure Variance
- Overhead Efficiency Variance

11.1 Overhead Expenditure Variance

This variance identifies the difference between the relevant standard/budgeted overhead expenditure and actual overhead expenditure.

Result

₹37,500 Favourable

11.2 Overhead Efficiency Variance

This variance measures the cost impact associated with the difference between standard activity hours and actual activity hours.

Result

₹62,500 Favourable

11.3 Total Overhead Variance

Component	Variance
Overhead Expenditure Variance	₹37,500 Favourable
Overhead Efficiency Variance	₹62,500 Favourable
Total Overhead Variance	₹1,00,000 Favourable

The favourable overhead variance partially offsets the adverse material and labour variances.

12. Overall Variance

The final model results are:

Metric	Result
Standard Cost	₹13,38,75,000
Actual Cost	₹13,71,40,000
Overall Variance	₹32,65,000 Adverse
Variance per Vehicle	₹3,265 Adverse
Reconciliation	CHECK PASSED

The actual cost is higher than the standard cost, producing an overall adverse variance of **₹32.65 lakh**.

13. Variance Reconciliation

A reconciliation control is included in the model to confirm that the detailed component variances agree with the overall cost difference.

The model reports:

CHECK PASSED

This provides an internal control that helps confirm the consistency of the variance schedules and overall variance calculation.

14. Variance Summary

The major cost-category results are:

Cost Category	Variance
Material	₹30,80,000 Adverse
Labour	₹2,85,000 Adverse
Overhead	₹1,00,000 Favourable
Overall Variance	₹32,65,000 Adverse

The favourable overhead variance partially offsets the adverse material and labour variances.

15. Key Findings

Finding 1 — Material cost is the main driver

Material cost produces the largest adverse variance at:

₹30.80 lakh Adverse

Within material variance, the price variance of:

₹25.30 lakh Adverse

is the largest individual variance in the model.

Finding 2 — Labour rate is adverse but efficiency is favourable

Labour rate variance is:

₹4.35 lakh Adverse

while labour efficiency variance is:

₹1.50 lakh Favourable

The efficiency benefit therefore partially offsets the higher labour rate, resulting in:

₹2.85 lakh Adverse Labour Cost Variance

Finding 3 — Overhead performance is favourable

Both overhead components are favourable:

- ₹37,500 favourable expenditure variance
- ₹62,500 favourable efficiency variance

This results in:

₹1.00 lakh Favourable Total Overhead Variance

Finding 4 — Overall cost is above standard

The final actual cost exceeds the standard cost by:

₹32.65 lakh

This represents:

₹3,265 Adverse per vehicle

under the illustrative model.

16. Company Context

The project uses publicly available FY2025–26 information relating to Maruti Suzuki India Limited.

The contextual figures include:

Metric	FY2025–26
Production	Over 23.4 lakh vehicles
Total Sales	2,422,713 vehicles
Net Sales	₹1,743,695 million
Net Profit	₹144,454 million

These figures are used to provide context regarding the scale of the company and its automobile manufacturing operations.

They are not used as a substitute for internal product-level costing data.

17. Source Classification

The project uses three levels of information.

A. Publicly Reported Company Data

Public company information is used for contextual reference.

Examples include:

- Production
- Total sales
- Net sales
- Net profit

B. Industry / Benchmark-Informed Information

Government of India / Ministry of Steel / JPC information is used as benchmark context for steel pricing.

This information helps provide an external reference when developing illustrative assumptions.

C. Portfolio Modelling Assumptions

Detailed standard and actual cost inputs are illustrative assumptions created for this project.

These include:

- Material quantities
- Material prices
- Labour hours
- Labour rates
- Overhead amounts
- Standard costs
- Actual costs

These inputs should not be interpreted as actual Maruti Suzuki internal data.

18. Steel Quantity Assumption

The steel quantity used in the model is an **industry-informed illustrative assumption**.

Vehicle weight is used only as contextual reference when developing the assumption.

The model does not claim that the selected steel quantity represents Maruti Suzuki's:

- Actual steel consumption
- Actual bill of materials
- Actual material yield
- Actual procurement quantity
- Actual production standard

This distinction is intentionally maintained to prevent the illustrative model from being presented as actual company data.

19. Dashboard

The workbook includes a dedicated dashboard designed to provide a high-level management view of the analysis.

The dashboard contains:

- Standard Cost KPI
- Actual Cost KPI
- Overall Variance KPI
- Variance per Vehicle KPI
- Variance charts
- Key findings
- Reconciliation status

The dashboard is supported by the detailed calculations contained in the underlying worksheets.

20. Sign Convention

The model presents variance results using:

- **Favourable (F)**
- **Adverse (A)**

Favourable

A favourable variance indicates a cost outcome better than the relevant standard or benchmark.

Adverse

An adverse variance indicates a cost outcome worse than the relevant standard or benchmark.

The interpretation should always be considered together with the amount and the corresponding calculation.

21. Limitations

This project is intentionally simplified because it is a portfolio modelling exercise.

Illustrative Cost Data

The detailed standard and actual cost inputs are modelling assumptions rather than publicly reported company-level production costs.

Simplified Manufacturing Structure

Real automobile manufacturing involves considerably more detailed:

- Bills of materials
- Production stages
- Cost centres
- Labour categories
- Machine hours
- Overhead pools
- Procurement arrangements
- Inventory movements
- Production yields

The portfolio model simplifies these elements to demonstrate standard costing methodology.

Public Data Limitation

Public company disclosures generally do not provide the detailed product-level internal cost information required to reproduce a company's internal standard costing system.

22. Skills Demonstrated

Microsoft Excel

- Structured workbook design
- Formula-based calculations
- Financial modelling
- Variance calculations
- Dashboard development
- KPI cards
- Charts
- Reconciliation controls
- Data classification

Accounting & Finance

- Cost Accounting
- Standard Costing
- Variance Analysis
- Management Accounting
- Cost Control
- Financial Analysis
- Management Reporting
- KPI Analysis
- Cost Reconciliation

23. Final Disclaimer

This is an educational and portfolio modelling project. Maruti Suzuki India Limited is used as the company context only. The standard quantities, actual quantities, rates, costs and detailed variance inputs are illustrative modelling assumptions and are not confidential, internal, or officially reported Maruti Suzuki costing data. Publicly available company and industry information is used only to provide contextual reference and benchmark support.

24. Project Files

The repository contains the completed Excel model:

[Excel/](#)

[└─ Standard Costing Variance Analysis.xlsx](#)

The main project README provides the executive-level overview, while this document provides the detailed technical notes supporting the model.